MST 0441 – Workshop 2, practice paper

Problem 1

(a)

Cost of one task-unit with labor: w (one architect-hour gives one task-unit).
Cost of one task-unit with AI: r/A_i.

Within a task, an architect-hour and an AI-service unit produce the same task-unit at the same quality, so task output is $E_i + A_iK_i$ and one input can fully replace the other at the constant rate A_i.

The isoquant is a straight line ($\sigma = \infty$), so the office uses only the cheaper method in each task.

AI is used when it is strictly cheaper:

$$r/A_i < w \iff A_i > r/w = \bar{A}$$

With $w = 480$ and $r = 2400$: $\bar{A} = 2400/480 = 5$.

- Technical drawings: $10 > 5 \rightarrow$ AI
- Building-code checks: $8 > 5 \rightarrow$ AI
- Site reports: $5 = 5 \rightarrow$ labor (tie)
- Cost estimates: $5/2 < 5 \rightarrow$ labor
- Client meetings: $3/5 < 5 \rightarrow$ labor

Automated set = {technical drawings, building-code checks}.

(b)

$$\pi_i = \max\{0,\ 1 - r/(wA_i)\}$$

Technical drawings: $\pi = 1 - 2400/(480 \cdot 10) = 1 - 1/2 = 1/2$
Building-code checks: $\pi = 1 - 2400/(480 \cdot 8) = 1 - 5/8 = 3/8$
Site reports, cost estimates, client meetings: $\pi = 0$

$$G = (2/5)(1/2) + (1/5)(3/8) = 1/5 + 3/40 = 11/40 = 0.275$$

Saving = $0.275 \cdot 320\,000 =$ **88 000 kr**. The office pays $320\,000 - 88\,000 =$ **232 000 kr**.

(c)

New cutoff: $\bar{A} = 2400/600 = 4$. Tasks with $A_i > 4$: technical drawings (10), building-code checks (8) and site reports (5). Cost estimates (5/2) and client meetings (3/5) stay with labor.

Technical drawings: $\pi = 1 - 2400/6000 = 3/5$
Building-code checks: $\pi = 1 - 2400/4800 = 1/2$
Site reports: $\pi = 1 - 2400/3000 = 1/5$
Cost estimates, client meetings: $\pi = 0$

$$G = (2/5)(3/5) + (1/5)(1/2) + (3/20)(1/5) = 24/100 + 10/100 + 3/100 = 37/100 = 0.37$$

All-labor cost at the new wage: 320 000 · 600/480 = 400 000 kr.
Saving = 0.37 · 400 000 = **148 000 kr**. The office pays 400 000 − 148 000 = **252 000 kr**.

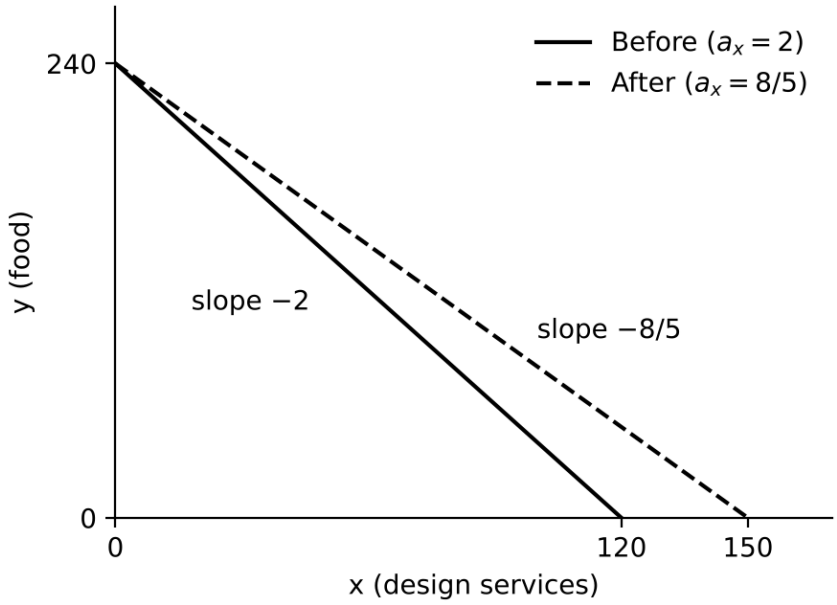
	w = 480	**w = 600**
G	0.275	0.37
Payment	232 000	252 000

G rises because the higher wage makes labor more expensive relative to AI, so the cutoff falls from 5 to 4, site reports switch to AI and every automated task saves a larger share of a higher labor cost. The payment still rises by 20 000 kr, because cost estimates and client meetings (1/4 of baseline costs) are still done by architects and become 25% more expensive. The AI tasks cost the same in kroner as before since r and A_i are unchanged, so the higher wage can only increase what the office pays.

(d)

PPF: $a_x x + a_y y = E$, so $y = E/a_y - (a_x/a_y)x$.

Before: $2x + y = 240$. x-intercept 240/2 = 120, y-intercept 240, slope −2.
After: $(8/5)x + y = 240$. x-intercept 240/(8/5) = 150, y-intercept 240, slope −8/5.



The frontier rotates outward around the y-intercept. Labor productivity in design services goes from 1/2 to 5/8 units per hour, so it rises by a factor (5/8)/(1/2) = **5/4** (25%).

The relative price equals the opportunity cost a_x/a_y only when the country produces both goods, so that labor earns the same wage in both industries (for example in autarky). If the country trades and specializes, the price of design services is set on the world market, so the frontier alone cannot tell us whether or how much the price falls. The frontier only shows what the economy as a whole can produce, not demand, prices or how income is split. In this one-factor model every worker earns

the same wage, so it cannot tell us who loses; for that we need a model that separates the architects whose tasks are automated from the owners of AI and a labor market that sets their wage.

Problem 2

(a)

Wage in ferries: p/a_x. Wage in sweaters: $1/a_y$. If both goods are produced the wages must be equal:

$$p/a_x = 1/a_y \implies p^{\mathrm{A}} = a_x/a_y$$

Fonna: $p^{\mathrm{A}} = 2/3$. Tarva: $p^{\mathrm{A}*} = 6/4 = 3/2$.

Absolute advantage: Fonna in both goods ($2 < 6$ and $3 < 4$).
Comparative advantage: Fonna in ferries ($2/3 < 3/2$), Tarva in sweaters (its opportunity cost of a sweater is $4/6 = 2/3$ ferry, against $3/2$ in Fonna).

(b)

Fonna produces only ferries: $x = E/a_x = 36/2 = 18$. Tarva produces only sweaters: $y^* = E^*/a_y^* = 108/4 = 27$.

Consumers spend 2/5 of income on ferries. World income is $pE/a_x + E^*/a_y^*$. Market clearing for ferries (spending = value of supply):

$$(2/5)(pE/a_x + E^*/a_y^*) = pE/a_x \implies (2/5)E^*/a_y^* = (3/5)pE/a_x \implies p = (2/3) \cdot (E^*/a_y^*) \,/\, (E/a_x)$$

$$p = (2/3) \cdot 27/18 = \mathbf{1}$$

$2/3 < 1 < 3/2$, so complete specialization holds.

Wages (in sweaters): Fonna $w = p/a_x = 1/2$. Tarva $w^* = 1/a_y^* = 1/4$.

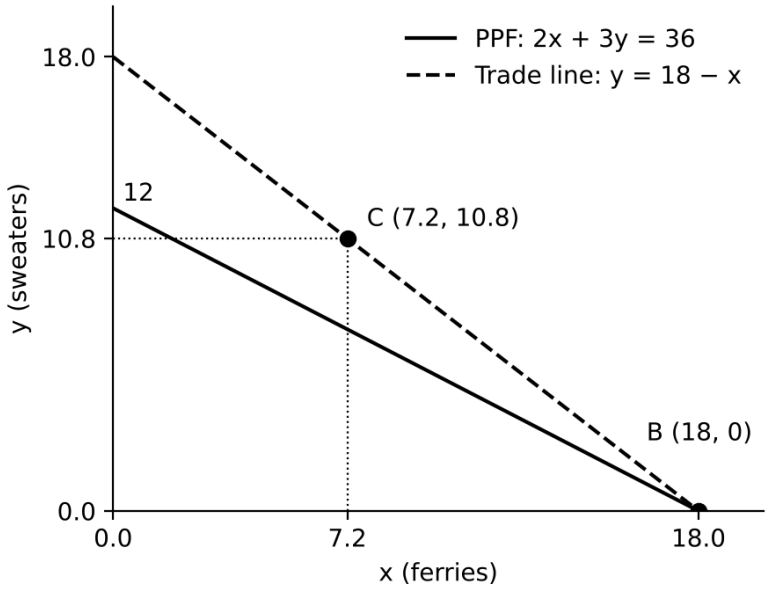
Unit cost	Ferries	Sweaters
Fonna	$2 \cdot 1/2 = 1$	$3 \cdot 1/2 = 1.5$
Tarva	$6 \cdot 1/4 = 1.5$	$4 \cdot 1/4 = 1$

Fonna is cheapest in ferries ($1 < 1.5$) and Tarva in sweaters ($1 < 1.5$), and the cheapest producer just covers its cost at the world price.

The pattern of trade is set by comparative advantage, that is relative labor requirements, not absolute productivity. The level of each wage is set by absolute productivity in the good the country produces valued at the world price ($w = p/a_x$, $w^* = 1/a_y^*$), which is why Fonna's wage is twice Tarva's.

(c)

Fonna's PPF: $2x + 3y = 36$, x-intercept 18, y-intercept 12, slope $-2/3$.
B = (18, 0). Trade line through B with slope $-p = -1$: $y = 18 - x$.
Income = 18 sweaters, so $x_C = (2/5) \cdot 18/1 = 7.2$ and $y_C = (3/5) \cdot 18 = 10.8$. C = (7.2, 10.8).



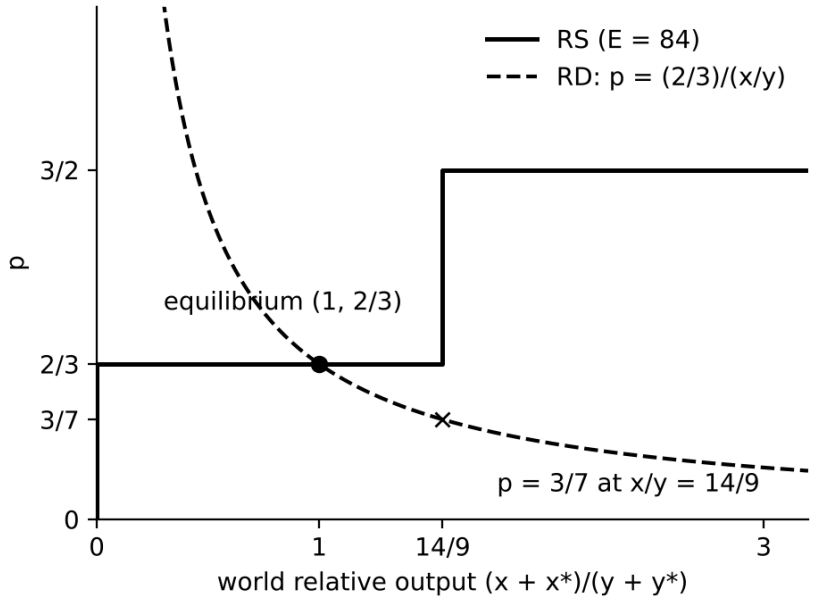
On the PPF at $x = 7.2$: $y = 12 - (2/3)\cdot 7.2 = 7.2 < 10.8$, so C lies above the frontier. Fonna exports $18 - 7.2 = 10.8$ ferries and imports 10.8 sweaters.

(d)

Now Fonna can make at most $84/2 = 42$ ferries. World relative supply (x/y):

- $p < 2/3$: only sweaters, x/y = 0
- $p = 2/3$: Fonna indifferent, x/y between 0 and $42/27 = 14/9$
- $2/3 < p < 3/2$: both specialize, x/y = 14/9
- $p = 3/2$: Tarva indifferent, x/y from 14/9 and up

Relative demand: spending shares give $px/y = 2/3$, so $x/y = (2/3)/p$.



With complete specialization x/y = 14/9, and RD then gives $p = (2/3)/(14/9) = 3/7$. Since $3/7 < 2/3$, Fonna would rather make sweaters, so complete specialization is not an equilibrium. RD instead

cuts RS on the flat step at $\boldsymbol{p = 2/3}$, where x/y = (2/3)/(2/3) = 1.

Tarva makes only sweaters: $y^* = 27$, $x^* = 0$. Fonna makes both, with $2x + 3y = 84$ and $x = y + 27$ (world x = world y):
$2(y + 27) + 3y = 84 \Longrightarrow 5y = 30 \Longrightarrow y = 6$, $x = 33$.

Fonna: **x = 33, y = 6**. Tarva: **x* = 0, y* = 27**.

Only Tarva gains, since it now buys ferries at $p = 2/3$, far below its autarky price of 3/2. Fonna gets no gain because the world price equals its own autarky price, so its wage (1/3 sweater) and consumption possibilities are the same as without trade. Fonna is so large that it must produce both goods, so its opportunity cost sets the world price and all the gains go to the smaller country.

Problem 3

(a)

Capital per unit of labor: boats 1/3, aluminium 2/1 = 2. Since 2 > 1/3, aluminium is capital-intensive and boats are labor-intensive.

Zero profit: $p_x = a_{Lx}w + a_{Kx}r, \quad p_y = a_{Ly}w + a_{Ky}r$

$$3w + r = 45 \qquad w + 2r = 40$$

From the first: $r = 45 - 3w$. Into the second: $w + 90 - 6w = 40 \Longrightarrow 5w = 50 \Longrightarrow \boldsymbol{w = 10}, \boldsymbol{r = 15}$.
Check: 30 + 15 = 45, 10 + 30 = 40.

(b)

Full employment:

$$\text{Labor: } 3x + y = 120 \qquad \text{Capital: } x + 2y = 80$$

$y = 120 - 3x \Longrightarrow x + 240 - 6x = 80 \Longrightarrow 5x = 160 \Longrightarrow \boldsymbol{x = 32}, \boldsymbol{y = 24}$.

With $E = 150$: $3x + y = 150$ and $x + 2y = 80$.
$y = 150 - 3x \Longrightarrow x + 300 - 6x = 80 \Longrightarrow 5x = 220 \Longrightarrow \boldsymbol{x = 44}, \boldsymbol{y = 18}$.

Changes: $\hat{x} = 12/32 = 37.5\%$, $\hat{E} = 30/120 = 25\%$, $\hat{K} = 0\%$, $\hat{y} = -6/24 = -25\%$.

$$\hat{x} = 37.5\% > \hat{E} = 25\% > \hat{K} = 0\% > \hat{y} = -25\%$$

This is the **Rybczynski theorem**.

w and r do not change (still 10 and 15). With goods prices fixed and both goods produced, the two zero-profit conditions alone pin down w and r independently of the endowments, so the extra labor is absorbed by shifting production toward the labor-intensive good (boats) rather than by a lower wage.

(c)

$$3w + r = 45 \qquad w + 2r = 50$$

$r = 45 - 3w \Longrightarrow w + 90 - 6w = 50 \Longrightarrow 5w = 40 \Longrightarrow \boldsymbol{w = 8}, \boldsymbol{r = 21}$.
Check: 24 + 21 = 45, 8 + 42 = 50.

Changes: $\hat{r} = 6/15 = 40\%$, $\hat{p}_y = 10/40 = 25\%$, $\hat{p}_x = 0\%$, $\hat{w} = -2/10 = -20\%$.

$$\hat{r} = 40\% > \hat{p}_y = 25\% > \hat{p}_x = 0\% > \hat{w} = -20\%$$

This is the **Stolper-Samuelson theorem**.

Workers lose in real terms, since the wage falls by 20% while no goods price falls, so the wage buys fewer boats and less aluminium. Capital owners gain, since the rental rate rises by 40%, more than either price (0% and 25%), so *r* buys more of both goods. Because the factor price changes lie outside the range of the goods price changes, each real return moves in the same direction measured in either good, so we need no information about what people consume.

Problem 4

(a)

Each firm has its own variety, so it faces a downward-sloping demand curve and can set price above marginal cost. Profits attract entry, which takes customers from every firm and shifts each firm's demand curve in until it just touches the average cost curve. There price equals average cost, which is above marginal cost because of the fixed cost, so the markup exactly covers the fixed cost and profit is zero.

(b)

When the markets are integrated, each firm competes with more varieties, so demand for each variety becomes more elastic and the markup falls. With a lower markup firms must sell more to cover the fixed cost, so the surviving firms expand, average cost and price fall, and the real wage w/p rises. Some firms exit, so the number of firms in each country falls, but consumers can buy the varieties from both countries, so the number of varieties available is still higher than in autarky. This connects to the evidence from the Canada–US Free Trade Agreement (Trefler), where tariff cuts led Canadian plants to exit, survivors to expand and labor productivity in Canadian manufacturing to rise.

(c)

Increasing returns (a fixed cost) mean each variety is made by only one firm in one country, so no country can produce all the varieties itself. Since consumers love variety, consumers in each country buy some of the other country's varieties, so even identical countries trade kayaks both ways (intra-industry trade).

Problem 5

1. c
2. a
3. b
4. d
5. c
6. d

7. a
8. b
9. a
10. c